

Code-Layout, Readability and Reusability

Date	04 July 2026
Team ID	SWTID-2026-2950
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Proper indentation is maintained throughout the project.	Yes	
2	Proper File Structure	Frontend, Backend, Database and API files are organized in separate folders.	Yes	
3	Meaningful Variable Names	Variables use meaningful names such as <code>loanAmount</code> , <code>monthlyIncome</code> , <code>settlementPrediction</code> .	Yes	
4	Function / Method Names	Functions are clearly named, e.g., <code>calculateFinancialHealth()</code> , <code>generateNegotiationLetter()</code>	Yes	
5	Code Comments	Important sections include comments for better understanding.	Yes	
6	Modular Design	Authentication, Loan Management, Dashboard and AI modules are separated.	Yes	
7	No Redundant Code	Duplicate code has been minimized using reusable functions.	Yes	
8	Error Handling	Input validation and API exception handling are implemented.	Yes	

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
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1	Authentication	FastAPI + JWT	Login & Registration	High
2	Loan Management Module	Python	Dashboard & Loan Section	High
3	Financial Calculator	Python	Financial Analysis	High
4	Gemini AI Module	Google Gemini API	Prediction & Negotiation Letter	High
5	Dashboard Components	React.js	Dashboard Pages	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project structure with separate frontend, backend, and database modules.
Readability	5	Code is easy to understand with meaningful naming conventions.
Reusability	5	Modules and functions are reusable across different project components.
Documentation / Comments	5	Sufficient comments and documentation are included.
Overall Score	5	Excellent code quality with maintainable and scalable architecture.